Problem Set 4

Ejercicio #1

1) $u(1 \circ a_1) = P_1 + 1 = 2$

$u(1 \circ a_2) = 1$

$u(1 \circ a_3) = 1 - P_3 = 0.$

Con orden $u(a_1) > u(a_2) > u(a_3)$.

2) a) $\forall \; P = (P_1, P_2, P_3)$ y $P' = (P_1', P_2', P_3')$

si hay función de utilidad, entonces.

$$u(P) \geq u(P') \quad ó \quad u(P') \geq u(P)$$

esto implica $P \succsim P'$ ó $P' \succsim P$ (Relación completa)

b) $\forall \; P, P', P'' \in \mathcal{L}$, si $u(P) \geq u(P')$ y $u(P') \geq u(P'')$

entonces $u(P) \geq u(P'')$.

Lo que implica: si $P \succsim P'$ y $P' \succsim P'' \Rightarrow P \succsim P''$

(Relación transitiva)

3) Continuidad : $\forall\ p \in \mathcal{L},\ \exists\ \alpha \in [0,1]$;

$$p \sim (\alpha \circ a_1 , (1-\alpha) \circ a_3)$$

Contra ejemplo:

$p = (0, 1/2, 1/2)$; $u(p) = -\frac{1}{2}$

$\exists\ \alpha$ s.t.

$$u(p) = \alpha u(a_1) + (1-\alpha) u(a_3)$$

$$-\frac{1}{2} = \alpha$$ X No es continua.

4) Tenemos que mostrar, $\forall\ \alpha, \beta \in [0,1]$.

$(\alpha \circ a_1, (1-\alpha) \circ a_3) \succsim (\beta \circ a_1, (1-\beta) \circ a_3)$ si y solo si $\alpha \geq \beta$.

$\iff \alpha u(a_1) + (1-\alpha) u(a_3) \geq \beta u(a_1) + (1-\beta) u(a_3)$

$$\alpha(1) \geq \beta(1)$$

$$\alpha \geq \beta \checkmark\checkmark$$ cumple con monotonicidad.

Ejercicio 2

$$u(x) = \frac{x^{1-\rho}}{1-\rho}$$

1) $E(u(x)) = P\,u(w - lw + x - xq) + (1-P)\,u(w - xq)$

2) $E(u(x)) = P\,\frac{(w(1-l) + x(1-q))^{1-\rho}}{1-\rho} + (1-p)\,\frac{(w - xq)^{1-\rho}}{1-\rho}$

C.P.O.

$$\frac{\partial E(u(x))}{\partial x} = P\,(w(1-l) + x(1-q))^{-\rho}(1-q) + (1-p)(w - xq)^{-\rho}(-q)$$

$$\frac{P}{1-P}\frac{(1-q)}{q} = \left(\frac{w(1-l) + x(1-q)}{w - xq}\right)^{\rho}$$

$$\underbrace{\left(\frac{P}{1-P}\,\frac{(1-q)}{q}\right)^{\frac{1}{\rho}}}_{A.} = \frac{w(1-l) + x(1-q)}{w - xq}$$

$$Aw - xAq = w(1-l) + x(1-q)$$

$$Aw - w(1-l) = x(1-q + Aq)$$

$$x^* = \frac{w(A - (1-l))}{1-q+Aq}$$

3) $\rho = q \Rightarrow A = 1 \Rightarrow K^{*} = w\ell.$

Ejercicio 3

1) $V(I) = P_H\, u\big(w - I + (1+r_H)I\big) + (1-P_H)\, u\big(w - I + (1+r_L)I\big)$

$$= P_H\, u(w + r_H I) + (1-P_H)\, u(w + r_L I)$$

Problema max:

$$\max_{I} \quad P_H\, u(w + r_H I) + (1-P_H)\, u(w + r_L I)$$

$$\text{s.a.} \qquad 0 \leq I \leq w.$$

2) $\max_{I} \; P_H\, u(w + r_H I) + (1-P_H)\, u(w + r_L I) + \lambda_1 (I) + \lambda_2 (w - I)$

C.P.O

$$P_H\, e^{-\alpha(w + I r_H)} (r_H) + (1-P_H)\, e^{-\alpha(w + r_L I)} r_L + \lambda_1 - \lambda_2$$

KTC.

$\lambda_1 I \geq 0$

$\lambda_2 (w - I) \geq 0.$

Si $I^* = 0. \Rightarrow \lambda_1 > 0$ y $\lambda_2 = 0.$

$$r_H P_H e^{-\alpha w} + (1-P_H)\, e^{-\alpha w} + \lambda_1 = 0.$$

$\lambda_1 > 0 \Rightarrow r_H P_H e^{-\alpha w} + (1-P_H) e^{-\alpha w} \leq 0.$

3) $I^* = w \Rightarrow \lambda_2 > 0$ y $\lambda_1 = 0.$

$\lambda_2 = 0 \Rightarrow P_H e^{-\alpha w(1 + r_H)} r_H + (1-P_H) r_L\, e^{-\alpha w(1 + r_L)} \geq 0$

4) $P_H \, e^{-\alpha(\omega + I r_H)}(r_H) + (1-P_H)\, e^{-\alpha(\omega + r_L I)} r_L = 0.$

$$r_H P_H \, e^{-\alpha I r_H} + (1-P_H)\, e^{-\alpha I r_L} r_L = 0.$$

$$r_H P_H \, e^{-\alpha I (r_H - r_L)} + (1-P_H) r_L = 0.$$

$$e^{\alpha I (r_L - r_H)} = \frac{-(1-P_H) r_L}{r_H P_H}$$

$$\alpha I (r_L - r_H) = \ln\left(\frac{-(1-P_H) r_L}{r_H P_H}\right) \Rightarrow \left(r_L < 0 \quad \text{sino, } I^* = \omega \right)$$

$$I^* = \frac{\ln\left(\frac{-(1-P_H) r_L}{r_H P_H}\right)}{\alpha (r_L - r_H)}$$